CyberPulse Setup Walkthrough

Full step-by-step guide to install, configure and run the Security Awareness Quiz Platform on your own machine. Covers repo access, Node, Supabase, database seeding, admin promotion, and testing with multiple user accounts. Includes a troubleshooting section for the problems we hit along the way.

Prerequisites

- A Linux VM or native Linux workstation with sudo access. Instructions assume Kali Linux running in UTM on a Mac, but any Debian/Ubuntu-based distro works.
- Collaborator access to the **CyArt-Tech/Security-Awareness-Quiz-Platform** GitHub repository. Ask the project owner if you don't have it.
- A GitHub account.
- A free Supabase account at supabase.com.
- About 45-60 minutes end-to-end. Budget extra time for the first run.

> **Architecture:** The whole stack runs locally. Only Supabase (the database) lives in the cloud. The VM/host, the app, and the browser all talk to the same Supabase project you create.

Part 1 - Install base tools

Open a terminal and install git, curl, Node.js 22, Mousepad (a lightweight text editor) and xclip (clipboard utility).

```
sudo apt update
sudo apt install -y git curl mousepad xclip
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo bash -
sudo apt install -y nodejs
node -v # should print v22.x or higher
```

Why Node 22: Supabase's SDK requires Node 22+. The default apt node package is often older. Use the NodeSource step above - don't skip it.

Part 2 - SSH key for GitHub

The repository is private. SSH keys are more reliable than tokens.

2.1 Generate the key

Press Enter at every prompt (no passphrase):

```
ssh-keygen -t ed25519
```

2.2 Copy the public key

```
cat ~/.ssh/id_ed25519.pub
```

Output is one long line starting with `ssh-ed25519`. Select the whole line, copy with **Ctrl+Shift+C**.

2.3 Add the key to GitHub

- Browser: github.com/settings/keys
- Click **New SSH key**.
- Title: anything (e.g. `work-vm`).
- Key type: **Authentication Key**.
- Paste the line into the Key box, click **Add SSH key**.

Access check: Confirm your GitHub account has access to the repo before proceeding. If git clone fails with 'Repository not found', the key is fine but the account isn't a collaborator.

Part 3 - Clone the repo and install dependencies

```
cd ~
git clone git@github.com:CyArt-Tech/Security-Awareness-Quiz-Platform.git
cd Security-Awareness-Quiz-Platform
npm install
```

The first time you git clone over SSH you may be asked *'Are you sure you want to continue connecting?'* - type **yes**, press Enter.

npm install pulls ~400 packages (1-2 minutes). Deprecation warnings are normal - ignore them. **Do not** run `npm audit fix --force` - it can break the project.

Part 4 - Create a Supabase project

4.1 Create the project

- Go to supabase.com, sign in (GitHub login works).
- Click **New project**.
- Name: anything (e.g. `cyberpulse`).
- Database password: pick one and **save it somewhere safe**.
- Region: closest to you.
- Click **Create new project**. Wait ~2 minutes until status is **Healthy**.

4.2 Collect the three credentials

Settings -> API Keys. Collect these three values:

- **Project URL** - looks like `https://<id>.supabase.co`.
- **Publishable key** - starts with `sb_publishable_`.
- **Secret key** - starts with `sb_secret_`. Click Reveal.

> **Legacy key names:** If you see 'anon' and 'service_role' keys instead of 'publishable' and 'secret', they're the same thing under the old names. Use anon for publishable, service_role for secret.

> **Security:** The secret key bypasses all database security. Never commit it to git, paste it in a chat, or share it in screenshots. Treat it like a password.

Part 5 - Configure environment variables

```
cd ~/Security-Awareness-Quiz-Platform
cp .env.example .env.local
nano .env.local
```

Fill in the three values from Part 4.2 - replace the placeholders:

```
NEXT_PUBLIC_SUPABASE_URL=https://YOUR-PROJECT-ID.supabase.co
NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY=sb_publishable_...
SUPABASE_SECRET_KEY=sb_secret_...
```

Save and exit nano: **Ctrl+O**, Enter, then **Ctrl+X**.

> **Formatting:** No spaces around **=**, no quotes around values. A single stray space will cause silent auth failures that are hard to debug later.

Part 6 - Initialize the database

The repo ships with `supabase/setup_complete.sql` - a single script that creates the schema, triggers, RLS policies, and seeds the 10 security domains, achievements and sample quizzes.

6.1 Open the SQL file

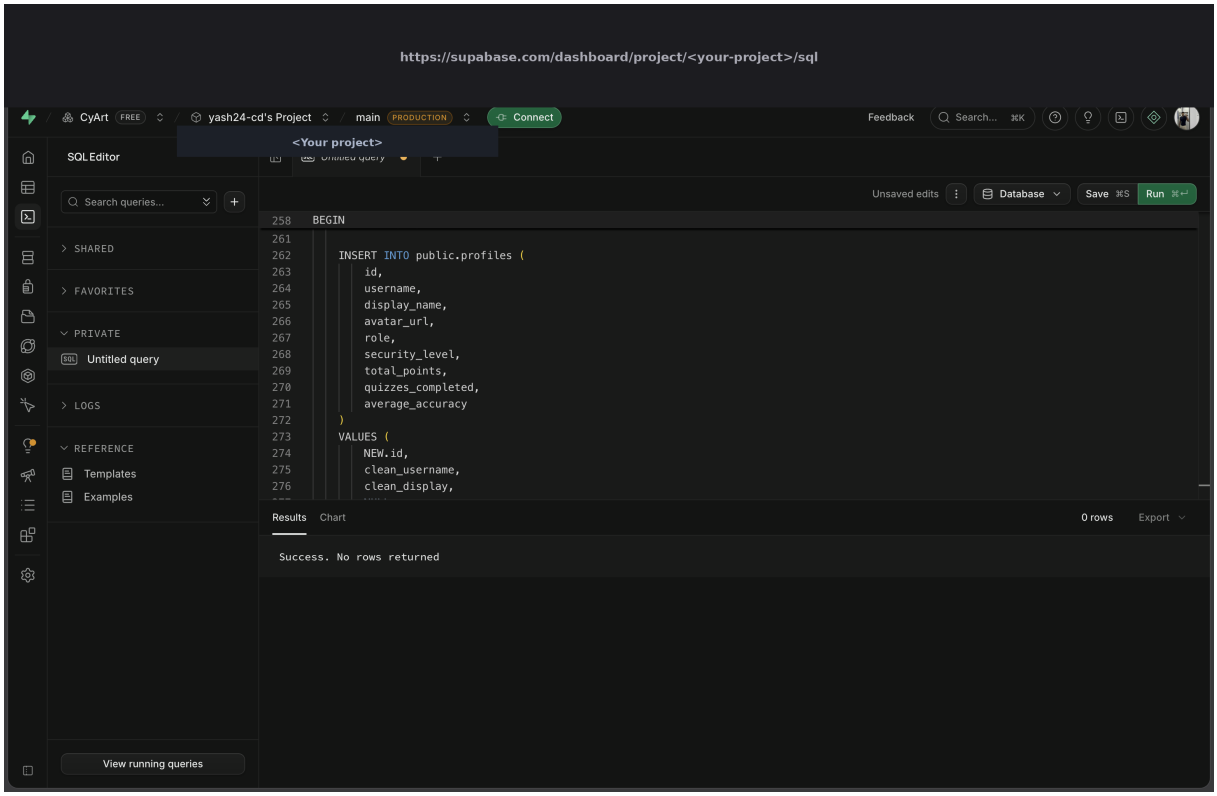
```
cd ~/Security-Awareness-Quiz-Platform
mousepad supabase/setup_complete.sql
```

6.2 Copy its contents

- Click inside the Mousepad text area.
- **Ctrl+A** to select everything.
- **Ctrl+C** to copy.
- Keep Mousepad open - the clipboard stays alive while it is.

6.3 Paste into Supabase SQL Editor

- In your browser, open your Supabase project.
- Left sidebar: **SQL Editor** -> **New query**.
- Click into the editor area, **Ctrl+V** to paste.
- Click the green **Run** button (bottom right) or press **Ctrl+Enter**.
- Supabase will warn about destructive operations - click **Run query** to confirm. This is expected: the script creates tables and triggers.



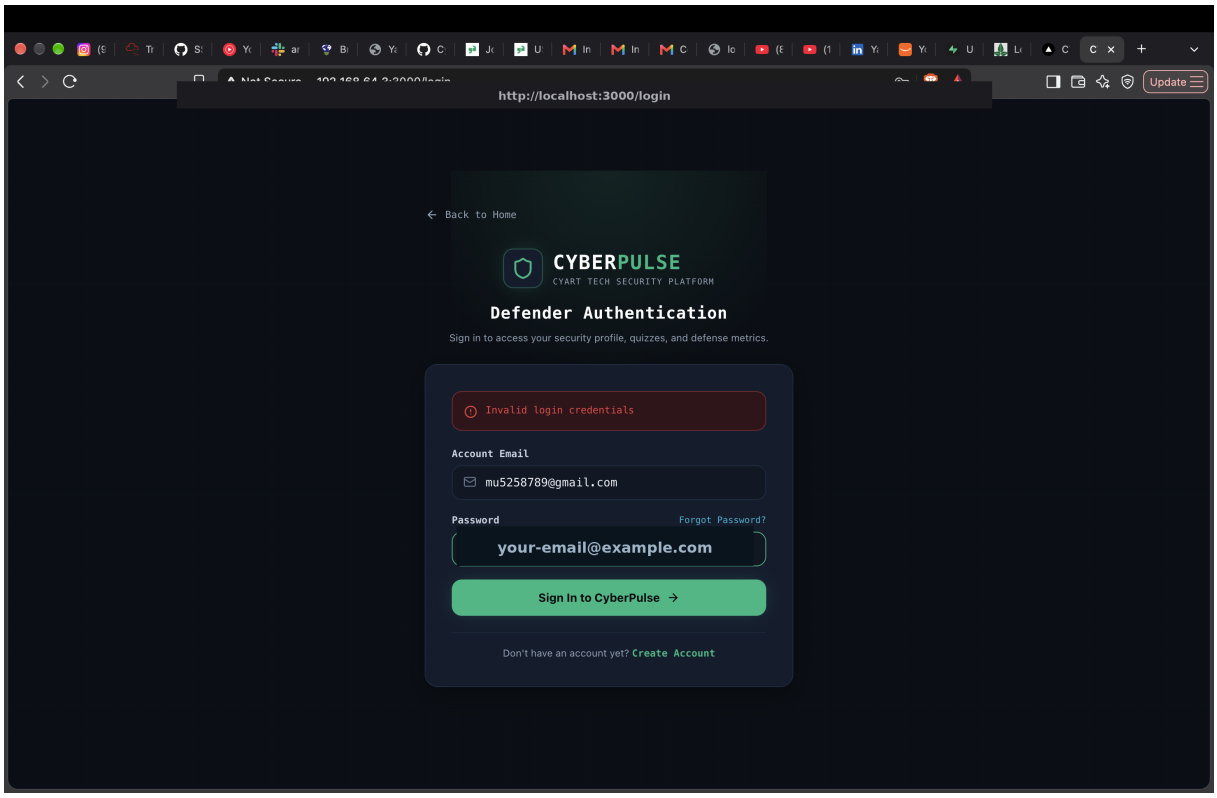
Expected result in the Supabase SQL Editor after running setup_complete.sql.

Expected output: Success. No rows returned is the right outcome. The script runs DDL (CREATE TABLE, CREATE FUNCTION, etc), not SELECT, so returning no rows is correct.

Part 7 - Start the dev server

```
cd ~/Security-Awareness-Quiz-Platform
npm run dev
```

Wait for 'Ready in ... ms' and the local URL to appear. Leave the terminal running - closing it stops the server.



The CyberPulse login page - this is what you'll see at http://localhost:3000

Part 8 - Create your admin account

8.1 Sign up through the app

- Browser: `http://localhost:3000`
- Click **Create Account** at the bottom.
- Enter an email and password. Save the password somewhere.
- You'll land on the dashboard. The 10 security domains should appear.

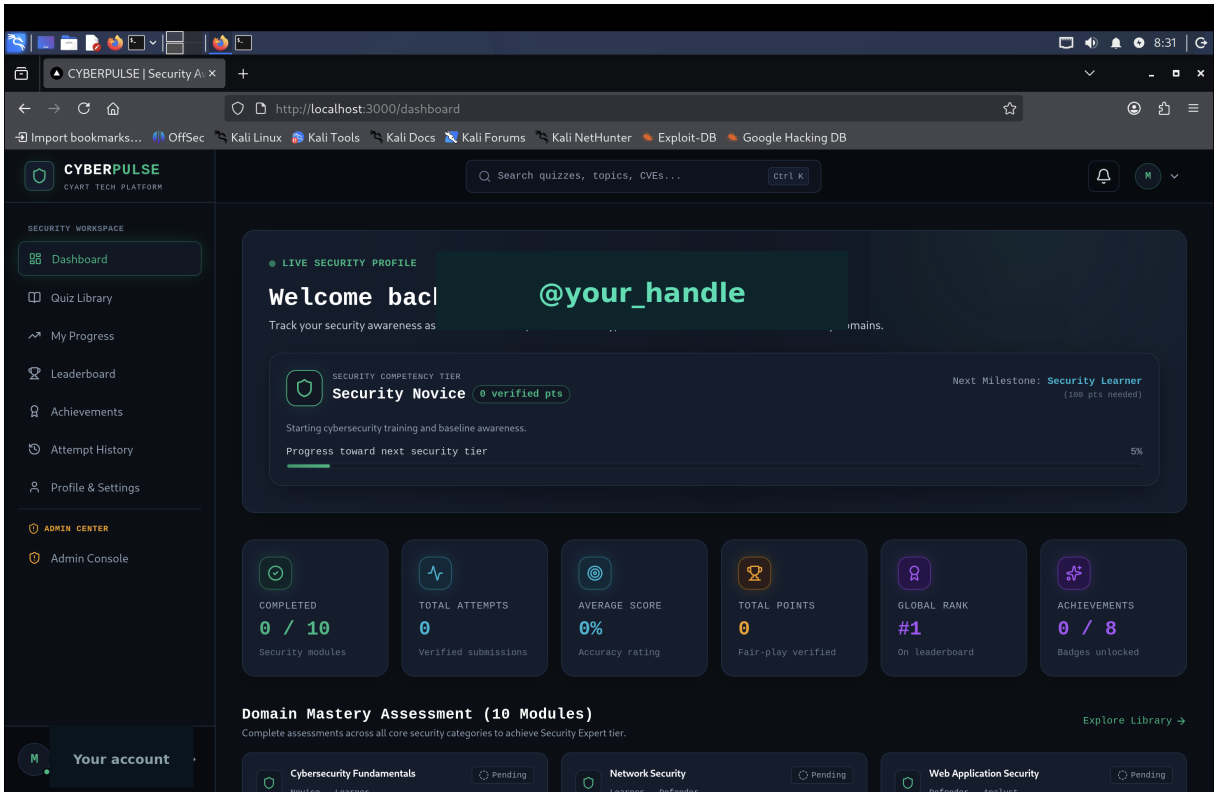
8.2 Promote the account to admin

By default the signup trigger assigns the `user` role. To unlock the Admin Center, update the role in Supabase.

- Supabase -> left sidebar -> **Table Editor**.
- Open the **profiles** table. Find your row (one row per signup).
- Double-click the **role** cell. Change `user` to `admin`, press Enter.

8.3 Sign out and back in

In the app, click your avatar (bottom-left of sidebar), then the logout icon. Sign in again with the same email and password. The role change takes effect on the new session.



The CyberPulse dashboard after signing in. Note the Admin Center section in the sidebar.

Part 9 - Test with a second (regular) user

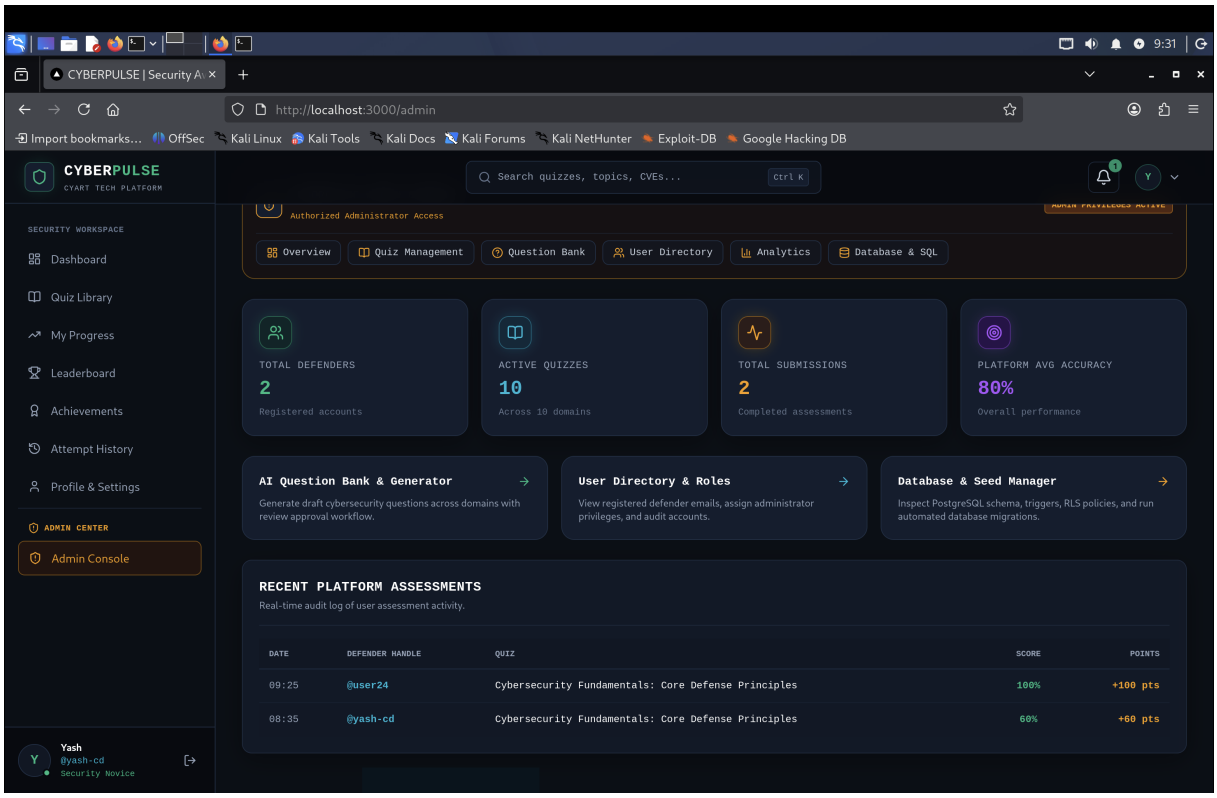
Confirm that non-admin users get the correct experience, and that admins can see them.

9.1 Create a regular user

- Open a private/incognito browser window (Ctrl+Shift+P in Firefox, Ctrl+Shift+N in Chrome).
- Go to `http://localhost:3000` and **Create Account** with a different email.
- The dashboard should show NO Admin Center. The sidebar should have only Dashboard, Quiz Library, My Progress, Leaderboard, Achievements, Attempt History, Profile & Settings.
- Take a quiz as this user to generate activity.

9.2 Verify from the admin side

- Back in your admin window, click **Admin Console**.
- You should see Total Defenders go up, and the regular user's quiz attempt appear in Recent Platform Assessments.



The admin console showing user stats and recent assessment activity.

Part 10 - Access the app from your host machine

When running inside a VM, Firefox-on-VM can get sluggish on Apple Silicon under UTM (and some other virtualization stacks). The cleanest workaround: run the dev server inside the VM, but access it from your host browser.

10.1 Find the VM's IP

```
ip a | grep 'inet 192'
```

You'll see a line like `inet 192.168.64.3/24 ...`. The IP is `192.168.64.3`.

10.2 Open from your host browser

On your host machine (Mac/Windows), open any browser and go to:

```
http://<VM-IP>:3000
```

Example: `http://192.168.64.3:3000`. Login, dashboard and everything else work the same way, with no VM-rendering slowdown.

> **IP may change:** The IP may change if the VM reboots. Re-run `ip a | grep 'inet 192'` after each reboot.

Troubleshooting

Problems encountered during the first setup, and the fixes that worked. If you hit one of these, jump to it.

A. git clone fails with 'Repository not found'

Your GitHub account isn't a collaborator on the repo, or the SSH key wasn't added correctly. Verify `cat ~/.ssh/id_ed25519.pub` matches what's listed at github.com/settings/keys, and that the repo owner has added you.

B. npm install prints engine warnings about Node version

Supabase packages require Node 22+. If `node -v` shows v18 or v20, re-run the NodeSource step from Part 1, then:

```
rm -rf node_modules
npm install
```

C. App loads but Dashboard says 'Welcome back, @' with no username

You signed up before the schema was in place, so the profile-creation trigger didn't fire. The signup created a row in `auth.users` but no `profiles` row. Fix by running this in Supabase SQL Editor:

```
INSERT INTO public.profiles (id, username, display_name, role)
SELECT id,
'user_' || substring(id::text, 1, 8),
split_part(email, '@', 1),
'admin'
FROM auth.users
WHERE id NOT IN (SELECT id FROM public.profiles);
```

Then sign out and sign back in.

D. 'Access Restricted' when clicking Admin Console

Your role is still `user`. Open Supabase -> Table Editor -> `profiles`. Find your row. Change `role` to `admin`. **Sign out and sign back in** - the change applies on the new session.

E. Firefox freezes the whole VM on page load

Common under UTM on Apple Silicon. Three mitigations, in order of least disruption:

- Close every other app in the VM. Keep only one terminal and Firefox open.
- Give Firefox a full 15-30 seconds after each click before clicking again. Repeated clicks make it worse.
- **Best fix:** run the dev server inside the VM but access the app from your host machine's browser (see Part 10).

> **Hard freeze recovery:** If the VM is completely frozen, force-stop it from UTM's main window (Stop button, Force stop) and restart. Your data on disk is safe - the Supabase database lives in the cloud and the project files are already saved.

F. xclip appears to copy, but paste gives nothing

xclip can lose large clipboard content when the pipe terminates. Use Mousepad instead: open the file in a GUI editor, Ctrl+A to select, Ctrl+C to copy, and keep the Mousepad window open while you paste.

G. 'Invalid login credentials' when you're sure the password is right

Three causes, in order of likelihood:

- **Browser autofill filled the wrong password.** Click the password field, Ctrl+A, Delete, and type it manually.
- **Caps Lock is on.** Check the keyboard indicator.
- **You forgot the password.** Reset it via Supabase SQL Editor:

```
UPDATE auth.users
SET encrypted_password = crypt('YOUR_NEW_PASSWORD', gen_salt('bf'))
WHERE email = 'your-email@example.com';
```

H. The 'potential issue detected' warning when running SQL

Supabase flags any schema-changing SQL (CREATE TABLE, DROP, etc) as destructive. This is expected for the setup script. Click the yellow **Run query** button to confirm.

I. Port 3000 is already in use

An old `npm run dev` is still running. Kill it and restart:

```
pkill -f 'next dev'
npm run dev
```

J. '.env.local' changes aren't picked up

Next.js reads env vars at startup. After any change to `.env.local`, stop the dev server with Ctrl+C and run `npm run dev` again. Also check for stray spaces around the = signs - a space breaks the parse silently.

K. Supabase SQL Editor shows 0 rows updated when you expected changes

The row you wanted to update may not exist, or the WHERE clause didn't match. Run a SELECT first to verify. Example:

```
SELECT u.email, p.role
FROM auth.users u
LEFT JOIN public.profiles p ON p.id = u.id;
```

What next

At this point you have:

- The app running locally.
- An admin account that can see the Admin Console, user directory, and analytics.
- A regular user account (optional, from Part 9) that can take quizzes and show up on the leaderboard.
- Full access to the Supabase project for schema changes and user management.

Common next steps

- Explore the Admin Console: Overview, Quiz Management, Question Bank, User Directory, Analytics, Database & SQL.
- Promote other users to admin through Admin Console -> User Directory & Roles (no SQL needed).
- Try the AI Question Bank & Generator for draft content.
- If you plan to share the app with your team, deploy to Vercel (needs the repo owner's approval first).

Known production hardening tasks

The repo README calls these out explicitly. Before any production deployment:

- Restrict or remove the `/api/setup-db` route - it can reset the schema.
- Lock down the admin SQL tools in the Admin Console.
- Add a proper RLS policy review.
- Add automated tests - the README states there are none.
- Introduce versioned migrations instead of the single `setup_complete.sql`.

Credits

CyberPulse is an internal CyArt Tech LLP project. The initial build was authored by Md Amir Ashraf as a capstone project. This walkthrough documents the first-time setup as experienced by onboarding team members.